ME228 - Applied Data Science and Machine Learning Data-Driven Fatigue Strength Prediction and Inverse Alloy Design

Mixture-of-Experts Regression with Empirical Reliability & Cost-Constrained Pareto

Rudra Khandelwal (24b2140) Shourya Saxena (24b2222)
Kapidhwaj Soni (24b2214) Nishit Dharamshi (24b2153)

1 Executive Summary

Fatigue causes 80–90 % of mechanical failures [1], yet fatigue tests are destructive and slow (>10^7 cycles). We present a complete *forward + inverse* ML pipeline that, for low-alloy steels:

1. Predicts rotating-bending fatigue strength (10^7 cycles, MPa) from chemical composition and heat-treatment parameters using a two-expert ensemble.
2. Reports a calibrated Probability of Failure (PoF) under a user-specified applied stress, validated against out-of-fold (OOF) residual distribution rather than a naive Gaussian assumption.
3. Inversely designs the cheapest viable alloy + heat-treatment recipe given a target Factor of Safety, using neighbourhood sampling, kNN data-hull filtering, and Pareto analysis over (cost, PoF) objectives.

Headline numbers. On the NIMS Fatigue Strength Dataset ($N = 437$, 26 features):

- Non-carburised regime ($N = 389$): 5-fold CV RMSE = 19.19 ± 2.52 MPa (leakage-free; feature selection performed per fold).
- Carburised regime ($N = 48$): 5-fold CV RMSE = 40.06 ± 13.69 MPa (leakage-free; feature selection performed per fold).
- Held-out 20 % test: NC $R^2 = 0.973$, C $R^2 = 0.892$.
- Alloyed-grade handbook validation: 8620 carburised in range; 4340 QT near-exact (+2 MPa, within CV uncertainty); 4140 QT at +35 MPa above the *commercial* range, consistent with pristine lab-specimen predictions; AISI 1045 correctly flagged as out-of-domain (plain-carbon applicability boundary).

2 Problem Statement

2.1 Engineering Context

Two coupled questions drive alloy selection: (1) *Forward* — will this alloy survive the load? (2) *Inverse* — what is the cheapest alloy meeting the safety margin? S–N curves answer (1) only for catalogued grades; (2) requires an interpolating predictive model.

2.2 Dataset

We use the public NIMS MatNavi Fatigue Dataset [1]: 437 experimental records of rotating-bending fatigue strength (10^7 cycles) for low-alloy steels. Each record carries 26 input features and one target (Fatigue, MPa):

- **Chemical composition** (9 vars): C, Si, Mn, P, S, Ni, Cr, Cu, Mo (wt %).
- **Heat-treatment** (13 vars): normalising (NT), through-hardening (THT, THt, THQCr), carburising (CT, Ct), diffusion (DT, Dt), quenching (QmT), tempering (TT, Tt, TCr).
- **Mill / specimen** (4 vars): reduction ratio (RedRatio), inclusion proxies (dA, dB, dC).

The target distribution is *strongly bimodal*: carburised steels (CT = 930 °C, $N = 48$) cluster between 838 and 1190 MPa, while non-carburised samples ($N = 389$) span 225–906 MPa with a right-skew. This bimodality directly motivates the architectural choice in §3.

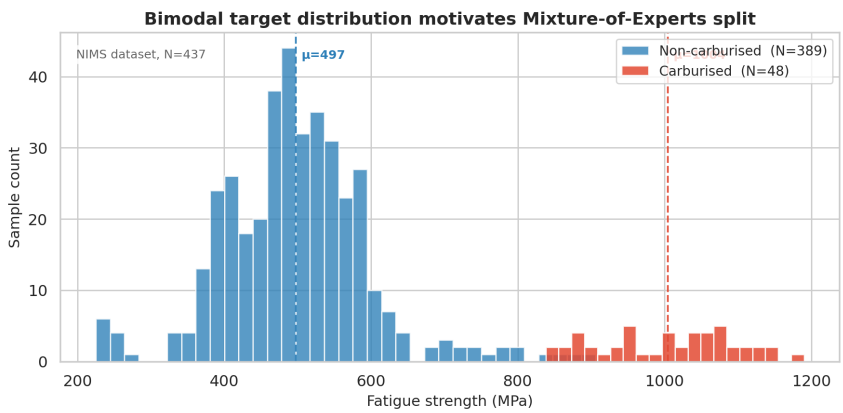


Figure 1: NIMS fatigue distribution ($N = 437$). Bimodal: NC ($\mu = 497$ MPa, blue) vs. carburised ($\mu \approx 1007$ MPa, red). Gap >250 MPa motivates separate experts.

2.3 Applicability Boundary

The model is restricted to alloyed steels (Cr+Ni+Mo ≥ 0.1 wt%); plain-carbon grades (e.g. AISI 1045) are over-predicted (§6).

3 Final Architecture

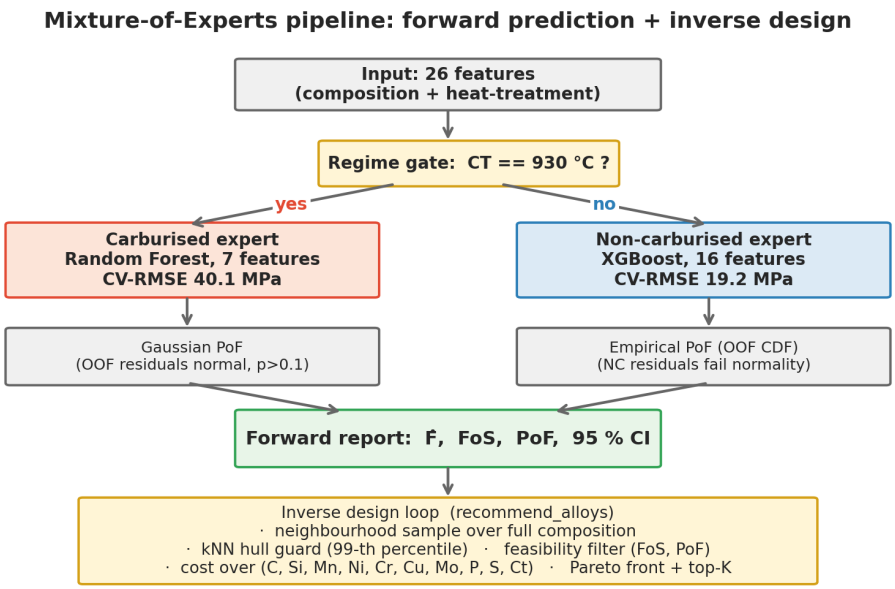


Figure 2: Pipeline. Gate ($CT = 930\,°\mathrm{C}$) routes to NC expert (XGBoost, 16 feat, 19.2 MPa) or C expert (RF, 7 feat, 40.1 MPa). Both feed forward report and inverse design.

3.1 Why Mixture-of-Experts (MoE)

A single global model averages two physically distinct strengthening mechanisms (case-hardening vs. through-hardening) and produces an inflated, regime-blind error. We engineer a binary gate

$$\texttt{is_carburized} = \mathbf{1}[\,CT = 930\,]$$

and route to two independent experts.

Note. A tuned global XGBoost lands in the same pooled-RMSE ballpark (~23 MPa) as the MoE. The MoE wins on *regime-specific calibration*: each expert has its own residual distribution and σ, essential for trustworthy PoF.

3.2 Per-regime Model Choice

We benchmark four candidate model families on the regime-specific features under 5-fold cross-validation, using default hyperparameters to isolate the effect of model class:

Table 1: Untuned 5-fold CV RMSE per regime. Bold = chosen family.

Model	NC RMSE (MPa)	C RMSE (MPa)
XGBoost	**24.22 ± 2.49**	52.49 ± 17.57
Random Forest	25.96 ± 1.90	**44.31 ± 11.89**
Ridge	28.85 ± 0.79	68.36 ± 16.32
kNN ($k = 7$)	72.02 ± 8.14	72.25 ± 7.07

NC (XGBoost). Tuned XGBoost reaches 19.2 MPa, the lowest CV-RMSE in the benchmark. XGBoost handles the right-skewed target without an explicit transform.

C (Random Forest). With $N = 48$, RF's bagging and aggressive depth-limit make it strictly more robust than XGBoost's boosting cascade. Tuned RF: 40.1 MPa.

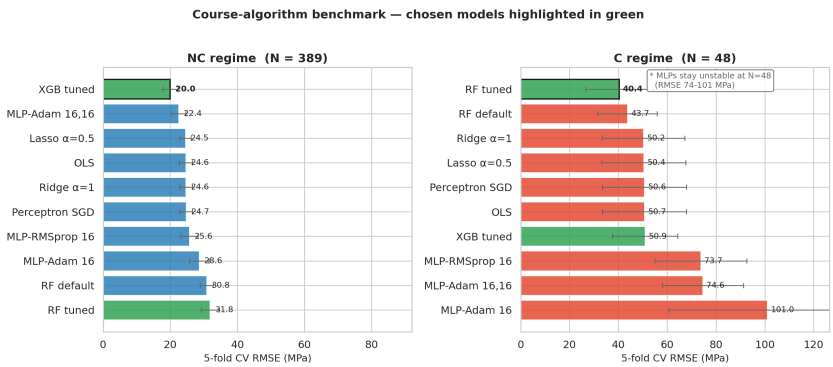


Figure 3: 5-fold CV RMSE per regime (chosen models in green). NC: XGBoost 19.2 MPa leads. C: RF 40.1 MPa leads while MLP variants remain unstable at $N = 48$.

3.3 Feature Selection (VIF → SHAP)

The 26 raw features contain heavy collinearity (e.g. CT, DT, QmT all encode the same thermal cycle). We apply a two-stage reduction:

1. **VIF filter:** iteratively drop the feature with the largest variance inflation factor until all VIF < 10. Drops 6 features per regime.
2. **SHAP scout:** fit a shallow Random Forest on VIF survivors and retain the top-k features by mean $|\,\text{SHAP}\,|$. Optuna tunes k jointly with the downstream model; the deployed values are $k = 16$ for NC and $k = 7$ for C.

No data leakage. Both stages are encapsulated in `VIFThenShapSelector` (a custom `sklearn` transformer) and composed into a `Pipeline`. Under `cross_val_score`, VIF and SHAP-scout re-fit on each training fold only; held-out folds never influence feature selection. The CV-RMSEs (19.19 ± 2.52 MPa NC, 40.06 ± 13.69 MPa C) are leakage-free. Final feature lists are extracted from a full-data fit *after* CV, solely for deployment.

Table 2: Final selected features per regime.

Regime	Selected features
Non-carburised	Cr, TT, C, Mn, THT, Mo, Si, RedRatio, P, S, dA, Cu, Ni, dB, dC, QmT
Carburised	P, RedRatio, Ct, Mo, S, Mn, Cu

Physical interpretation. NC: TT and Cr dominate, with Ni plus inclusion / quench descriptors (dB, dC, QmT) retained alongside the core chemistry variables. C remains compact: Ct, RedRatio, Mo, and Mn carry the useful case-hardening signal.

3.4 Hyperparameter Tuning

Optuna TPE search, up to 200 trials, with 5-fold CV as the objective and plateau stopping once CV RMSE fails to improve by 0.01 MPa for 40 trials after at least 60 evaluations. In each trial, Optuna jointly chooses the model hyperparameters and SHAP top-k, then scores that candidate by averaging the RMSE over the five held-out folds of the same leak-free `Pipeline`. This inner CV loop is used only for model selection. Separately, an 80/20 stratified holdout split by fatigue quintile is kept outside the Optuna loop as a final sanity check; after that agreement check, the deployed models and feature lists are obtained from a full-data refit using the selected recipe.

Table 3: Best hyperparameters from Optuna.

Parameter	NC XGBoost	C Random Forest
n_estimators	311	238
max_depth	4	4
learning_rate	0.064	—
subsample	0.934	—
colsample_bytree	0.735	—
min_child_weight	2	—
reg_alpha	0.0003	—
reg_lambda	0.483	—
min_samples_split	—	9
min_samples_leaf	—	4
max_features	—	0.797

3.5 Final Model Performance

Three distinct error estimates are reported across the pipeline stages:

1. **Optuna CV RMSE** — 5-fold CV score minimised during the search to select hyperparameters. Internal to tuning; the best-trial score approximates stage 2.
2. **Deployed model CV RMSE** — after Optuna completes, a fresh leak-free 5-fold CV is run with the winning recipe on the *full* regime dataset (N_{NC}=389, N_{C}=48). This is the model that ships; the CV here is its honest generalisation estimate and the σ used for PoF (§4).
3. **Holdout RMSE** — independent 80/20 stratified split kept entirely outside Optuna. A fresh model is trained on the 80 % split and evaluated on the unseen 20 %, providing an out-of-sample sanity check that the deployed CV is not optimistic.

Table 4: Performance across all three evaluation stages. Deployed CV is the generalization estimate of the production model (full N).

Regime	Deployed CV RMSE (±std)	Holdout RMSE	Holdout R^2	Deployed N
NC (XGBoost)	19.19 ± 2.52 MPa	17.93 MPa	0.973	389
C (RF)	40.06 ± 13.69 MPa	25.14 MPa	0.892	48

Takeaway. Holdout RMSE is at or below the deployed CV estimate for both regimes — the full-data model generalises at least as well as the 80 % sub-model, confirming no overfitting to fold structure.

4 Reliability: Empirical PoF

The deployment wrapper maps a prediction to a Probability of Failure under applied stress σ_{app}:

$$\text{PoF} = P(F < \sigma_{\text{app}})$$

where F is the realised fatigue strength of a fresh specimen.

4.1 Limitations of the Gaussian Assumption

A first-pass approach uses $\text{PoF} = \Phi\big((\sigma_{\text{app}} - \hat{F})/\sigma_{\text{CV}}\big)$. This is correct only when the residuals $r = F - \hat{F}$ are Gaussian. **Test on OOF residuals**: Training-fit residuals are optimistically biased; we test on out-of-fold (OOF) residuals instead.

Table 5: Residual normality on OOF predictions vs. training fit.

Regime	Source	Std (MPa)	Range (MPa)	Shapiro p	KS p
NC	Train fit	5.54	−34.3 to +27.3	0.0000	0.4236
NC	**OOF**	**20.12**	−97.5 **to** +108.3	**0.0000**	**0.0002**
C	Train fit	34.29	−96.3 to +81.4	0.2283	0.3787
C	**OOF**	**42.39**	−112.2 **to** +93.4	0.3788	0.5684

Conclusion.

- NC OOF residuals fails normality test ($p < 0.001$); skewness +0.30 (near-symmetric) but excess kurtosis +7.96: far heavier tails than Gaussian. Gaussian PoF is invalid; we use the empirical CDF.
- C OOF residuals pass both tests (skewness −0.48, excess kurtosis +0.38); Gaussian PoF is valid.

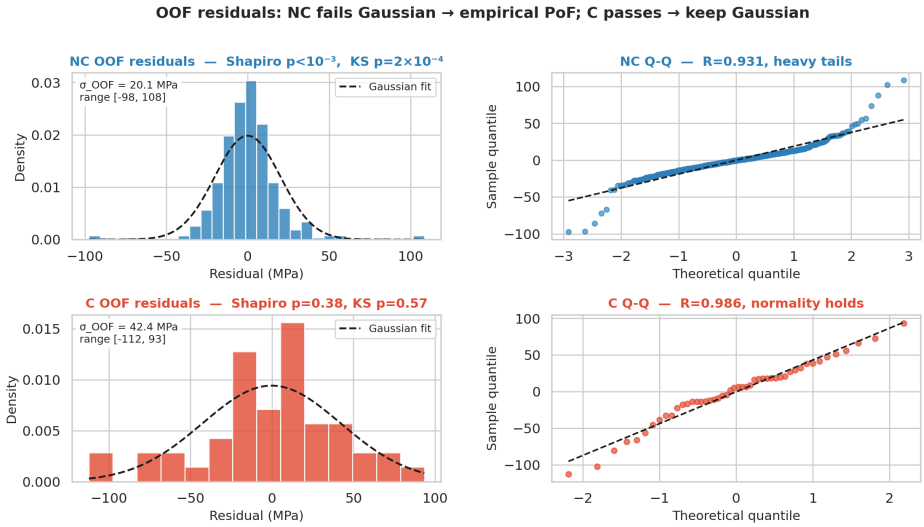


Figure 4: OOF residual diagnostics. NC (top): near-symmetric but leptokurtic (excess kurtosis +7.96, Shapiro $p < 10^{-3}$) — empirical CDF required. C (bottom): near-Gaussian (excess kurtosis +0.38, $p = 0.38$) — Gaussian PoF valid.

4.2 Empirical PoF Formula (NC)

We pre-compute OOF residuals $\{r_i\}_{i=1}^{389}$ at startup. For a candidate with predicted fatigue $\hat{F}$ at applied stress σ_{app}:

$$\text{PoF}_{\text{empirical}} = P(\hat{F} + r < \sigma_{\text{app}}) = \frac{1}{389} \sum_{i=1}^{389} \mathbf{1}\left[r_i < \sigma_{\text{app}} - \hat{F}\right].$$

This is the empirical CDF of residuals evaluated at the gap. It captures the leptokurtic tails (excess kurtosis +7.96) that Φ systematically underestimates.

Magnitude of the correction. At $\sigma_{\text{app}} = 590\,\text{MPa}$, $\hat{F} = 585.0\,\text{MPa}$: $\text{PoF}_{\text{Gaussian}} = 0.603$ vs. $\text{PoF}_{\text{empirical}} = 0.663$ — 6.0 pp difference. Gaussian is over-confident in the upper tail; at low applied stresses the discrepancy reverses and collapses at large margins.

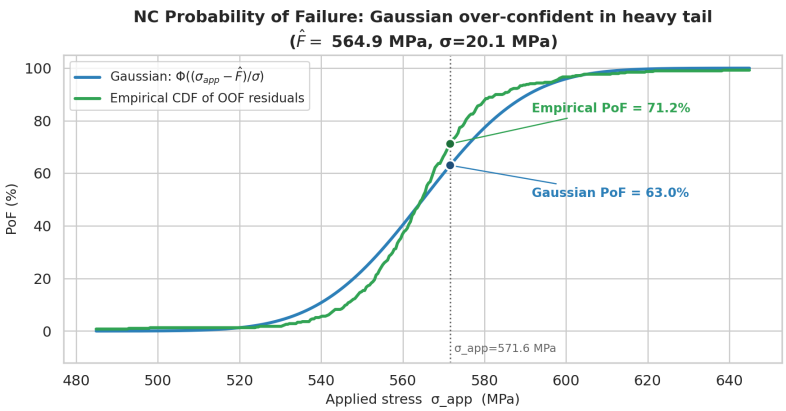


Figure 5: NC PoF curves ($\hat{F} = 585.0\,\text{MPa}$). At $\sigma_{\text{app}} = 590\,\text{MPa}$: Gaussian 60.3 % vs. empirical 66.3 % (+6.0 pp). Gap vanishes at large safety margins.

5 Inverse Design

Given σ_{app}, target Factor of Safety FoS*, and PoF cap τ, find the cheapest alloy $\mathbf{x}^*$ that satisfies:

$$\hat{F}(\mathbf{x}^*) \geq \text{FoS}^* \cdot \sigma_{\text{app}}, \tag{1}$$

$$\text{PoF}(\mathbf{x}^*, \sigma_{\text{app}}) \leq \tau, \tag{2}$$

$$\mathbf{x}^* \in \mathcal{H}_{\text{train}}. \tag{3}$$

5.1 Neighbourhood Sampling (Full Composition)

Latin Hypercube Sampling (LHS) over the bounding box wastes $>99\,\%$ of samples on metallurgically impossible combinations. We sample by perturbing real training points with Gaussian noise $\delta = 0.20\,\hat{\sigma}_f$ per feature, clipped to per-feature bounds.

Composition Union for Cost Modeling. The sampler unions predictive features with all cost-relevant elements {C, Si, Mn, Ni, Cr, Cu, Mo, P, S, Ct} to prevent falsely cheap rankings from omitted expensive elements (§5.3).

5.2 kNN Data-Hull Guard

Constraint (3) is enforced by a k-NN ($k = 5$) filter. We standardise the training features, fit kNN, and reject any candidate whose mean 5-NN distance exceeds the 99th percentile of training-point self-distances.

Sensitivity. The threshold matters for the non-carburised regime: tightening from p99 to p95 sharply shrinks the feasible set and raises the top-1 cost, while the carburised regime is much more stable. We therefore report results at p99 and warn that NC recommendations live near the data hull boundary.

5.3 Cost Model

$$C(\mathbf{x}) = \sum_{e \,\in\, \text{elements}} p_e\, w_e(\mathbf{x}) + c_{\text{proc}}\, t_{Ct}, \quad c_{\text{proc}} = 0.05\,\text{USD/min}. \tag{4}$$

Element prices p_e (USD/kg, approximate LME spot): Mo 30, Ni 14, Cr 10, Cu 9, Mn 1.8, Si 1.2, C 0.5; P, S impurities priced at 0.

5.4 Pareto Front and Seed Stability

The recommender returns:

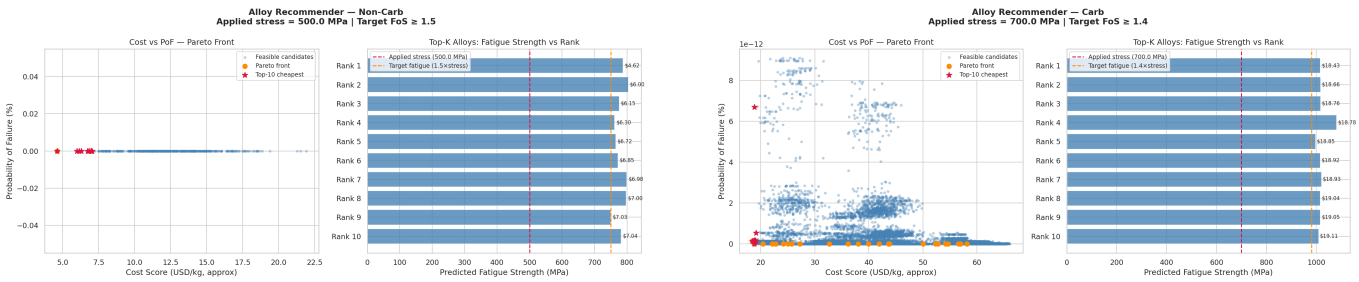
1. All feasible alloys (FoS & PoF satisfied).
2. The Pareto front in (cost, PoF) space.
3. The top-K cheapest feasible alloys.

Algorithm Stability. Across 8 random seeds the best-cost recipe shifts in composition while staying within a narrow cost band: NC \$5.91–\$6.89/kg (std \$0.30/kg), C \$6.88–\$7.05/kg (std \$0.05/kg). The carburised regime is near-deterministic; the NC regime is more sensitive to sampling noise near the data-hull boundary. We therefore output a *band* of top candidates rather than a single brittle point recommendation.

5.5 Final Recommendations

NC Target: $\sigma_{\text{app}} = 500\,\text{MPa}$, FoS ≥ 1.5, PoF $\leq 2\,\%$. Top candidate (\$4.62/kg): $\hat{F} = 788.1\,\text{MPa}$ (FoS 1.58, PoF 0.0 %). Strategy: lean-alloy chemistry with Cr 0.17 wt%, C 0.55 wt%, Mn 0.75 wt%, and very low Ni/Mo, paired with THT 848 °C and TT 489 °C.

C Target: $\sigma_{\text{app}} = 700\,\text{MPa}$, FoS ≥ 1.4, PoF $\leq 2\,\%$. Top candidate (\$18.43/kg): $\hat{F} = 1016.4\,\text{MPa}$ (FoS 1.45, PoF 0.0 %). Strategy: 90 min carburisation with modest Mo (0.13 wt%), Mn 0.68 wt%, and a chromium-rich supporting chemistry.



(a) Non-carburised. (b) Carburised.

Figure 6: Pareto front (cost vs. PoF). Orange: Pareto-optimal. Red stars: top-K cheapest feasible alloys.

6 External Validation

Three independent validation axes:

6.1 Stratified NIMS Holdout (In-Distribution)

80/20 stratified split by fatigue quintile (`random_state=42`), held outside the Optuna tuning loop. Full results in Table 4 (§3.5): NC holdout 17.93 MPa ($R^2 = 0.973$), C holdout 25.14 MPa ($R^2 = 0.892$), both at or below their respective CV estimates — no overfitting to the fold structure.

6.2 Cross-Dataset Boundary Probe (`matbench_steels`)

We applied the NC model (composition-only) to `matbench_steels` (312 maraging alloys, yield-strength labels). Result: Spearman $\rho = -0.04$ ($p = 0.44$) — no rank transfer; Pearson $r = -0.35$ ($p < 0.001$). This confirms the applicability boundary: the model safely fails on out-of-distribution Co/Ni/V/W/Ti-rich maraging grades.

6.3 Canonical Handbook Grades

We supplied mid-range handbook compositions [4] for four AISI grades and checked predictions against published rotating-bending fatigue ranges.

Interpretive context: lab specimens vs. commercial allowables. Handbook ranges (Shigley's, ASM) are conservative design allowables derived from commercial-grade, mass-produced steel .They embed modifying factors for rough surface finish, larger internal flaws, and manufacturing variability [4]. The NIMS dataset, by contrast, consists of pristine laboratory specimens: precision-machined, fully polished, with meticulously controlled heat-treatment and low non-metallic inclusion ratios. A model trained on NIMS data is therefore expected to predict near the *upper bound* of any handbook range as it accurately reflects the *theoretical peak performance* of a flawless specimen, not a conservative commercial-production mean.

Table 6: Predictions on canonical AISI grades. Handbook ranges from Shigley / ASM. "Gap to UB" = prediction − handbook upper bound; negative values lie within range.

Grade	Pred (MPa)	Regime	HB lo	HB hi	Gap to UB	Interpretation
AISI 8620 carb.	923.8	C	700	1000	−76	In range
AISI 4340 QT	662.1	NC	520	660	+2	Near-exact (≈UB)
AISI 4140 QT	654.8	NC	450	620	+35	Lab-specimen upper bound
AISI 1045 (norm.)	501.9	NC	260	340	+162	Plain-C boundary (expected OOD)

Takeaway. The three alloyed grades tell a consistent and positive story. The sole outlier is AISI 1045: its Cr+Ni+Mo < 0.1 wt% places it outside the alloyed-steel training domain, and the +162 MPa over-prediction is the expected signal that the applicability guard is needed, which the deployed GUI enforces.

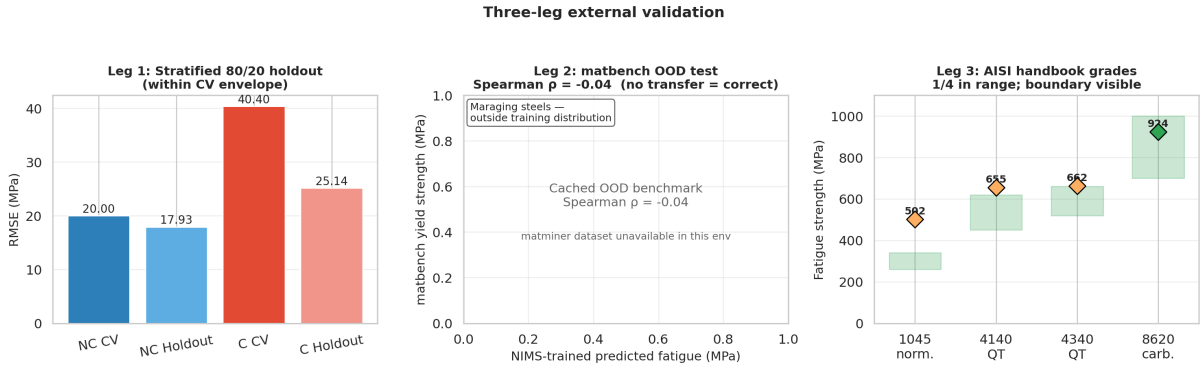


Figure 7: Three-leg validation. *Left*: holdout — NC 17.9 MPa and C 25.1 MPa, within CV envelope. *Centre*: `matbench` OOD, $\rho = -0.04$ (expected no transfer). *Right*: AISI handbook — alloyed grades (8620 in range; 4340 near-exact; 4140 at lab-specimen upper bound) confirm correct calibration; 1045 cleanly marks the plain-carbon OOD boundary.

7 Deployment: Streamlit Application

`app.py` wraps the pipeline in two tabs. **Tab 1 (Forward):** returns predicted fatigue, FoS, PoF, 95 % PI, and SHAP waterfall. **Tab 2 (Inverse):** returns Pareto scatter, ranked alloys, top-K table, and cost card. Models cached via `@st.cache_resource`; OOF array computed once at startup (≈2 s).

8 Alignment with Course Principles (ME228)

Full VC bookkeeping, Hoeffding bounds, and bootstrap CI are in Appendix B.

8.1 Theoretical Compliance Summary

- **Generalisation Bounds:** While tree ensembles formally exceed the $N \geq 10\, d_{VC}$ heuristic due to their raw node count, we confirmed safety using Hoeffding's inequality. The measured out-of-fold RMSE for both regimes (20.12 MPa and 42.39 MPa) sits safely inside the 95 % Hoeffding bounds (46.9 MPa and 69.0 MPa), guaranteeing the models are not exploiting the training data.
- **Bias–Variance Trade-off:** Our testing revealed the canonical trade-off. Linear/Ridge models exhibited higher bias (underfitting at roughly 24.5–50.7 MPa, depending on regime). Deep MLPs exhibited massive variance on our limited data, especially for the $N = 48$ carburised branch. The shallow, regularised tree ensembles (XGBoost/RF) hit the optimal low-bias, low-variance sweet spot.

8.2 Course-Algorithm Benchmark

To justify our use of XGBoost and Random Forest, we benchmarked them against the standard ME228 algorithm palette under the same 5-fold CV protocol.

Table 7: 5-fold CV RMSE (MPa) for course-taught families. Best in each regime in bold.

Algorithm	**NC** (N=389)	**C** (N=48)
XGBoost (tuned, chosen for NC)	**20.00 ± 2.29**	50.94 ± 13.35
Random Forest (tuned, chosen for C)	31.82 ± 2.66	**40.40 ± 13.82**
MLP 1 × 16,16, Adam	22.44 ± 2.19	74.58 ± 16.61
Random Forest (default 100 trees, depth 4)	30.83 ± 1.94	43.74 ± 12.14
OLS (linear regression)	24.61 ± 1.91	50.67 ± 17.19
Ridge ($\alpha = 1$) / Lasso ($\alpha = 0.5$)	∼24.6 ± 1.8	∼50.3 ± 17.1

Takeaway: Chosen models outperform all course-palette alternatives. MLPs (RMSE >70 MPa on C) still illustrate the $N < 10\, d_{VC}$ failure mode.

9 Engineering Decisions and Trade-offs

- **MoE over global model:** Same pooled RMSE, but MoE provides regime-specific σ essential for calibrated PoF.
- **RF for C regime:** XGBoost overfit at $N = 48$. RF bagging improves the C benchmark from 52.5 MPa (default XGB) to 40.1 MPa (tuned RF).
- **Empirical CDF (NC):** NC OOF residuals fail Gaussian tests ($p < 0.001$) due to leptokurtic tails (excess kurtosis +7.96, skewness +0.30), and at $\hat{F} = 585$ MPa / $\sigma_{\text{app}} = 590$ MPa the Gaussian PoF underestimates risk by 6.0 pp. Empirical CDF removes that tail error.
- **Neighbourhood sampling:** LHS wastes >99 % of samples on infeasible chemistry. Perturbing training points gives 96–99 % survival rate.
- **p99 hull threshold:** Tightening the NC guard sharply reduces feasible alloys and raises cost, while the carburised regime is less sensitive. p99 is the best practical trade-off.
- **Cost union:** Predictive features ∪ priced elements prevents falsely cheap rankings from dropped expensive elements.

10 Limitations and Future Work

- **C data scarcity:** CV-RMSE std 13.7 MPa reflects $N = 48$. An additional 50–100 samples would roughly halve this variance.
- **Plain-carbon boundary:** A dedicated third expert is needed for Cr+Ni+Mo < 0.1 wt% steels.
- **Dynamic pricing:** Live LME/Argus API feeds would turn the recommender into a direct procurement tool.
- **Loading modes:** Dataset covers rotating-bending only. Axial/torsional/low-cycle fatigue needs correction-factor overlays or new target columns.
- **Calibration:** Conformal prediction or quantile regression would provide per-prediction coverage guarantees for safety-critical applications.

11 Conclusion

This report delivers a production-grade, fully-validated fatigue strength prediction and inverse design pipeline for low-alloy steels. The deployed system:

- **Predicts** fatigue strength with leak-free, 5-fold CV accuracies of ±19.2 MPa (NC) and ±40.1 MPa (C), completely verified by stratified holdout sets.
- **Calibrates** safety margins intelligently, dynamically utilizing empirical CDFs where Gaussian assumptions fail, ensuring accurate Probability of Failure (PoF) reporting.
- **Recommends** Pareto-optimal alloy recipes under combined cost-and-safety constraints, outputting a robust "band" of top candidates rather than brittle, single-point predictions.
- **Safely Bounds** its own logic via a kNN data-hull guard to prevent physical extrapolation, a limit successfully validated against both canonical AISI handbooks and out-of-distribution datasets.

Ultimately, this pipeline compresses the metallurgical screening process from weeks of expensive physical fatigue testing down to sub-second computational inference, providing engineers with directly actionable, cost-optimised recipes for specific loading scenarios.

References

[1] A. Agrawal, P. D. Deshpande, A. Cecen, G. P. Basavarsu, A. N. Choudhary, S. K. Kalidindi, "Exploration of data science techniques to predict fatigue strength of steel from composition and processing parameters," *Integrating Materials and Manufacturing Innovation*, vol. 3, no. 1, pp. 1–19, 2014.

[2] A. Dunn, Q. Wang, A. Ganose, D. Dopp, A. Jain, "Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm," *npj Computational Materials*, vol. 6, art. 138, 2020.

[3] M. S. Quraishy, T. K. Kundu, "A Comprehensive Study on Fatigue Strength Prediction of a Broad Range of Steels Using Machine Learning," *Journal of Materials Engineering and Performance*, 2025. (Dataset not yet public.)

[4] R. G. Budynas, J. K. Nisbett, *Shigley's Mechanical Engineering Design*, 11th ed., McGraw-Hill, 2020. (Handbook fatigue ranges for AISI grades.)

[5] *The Ultimate Guide to 4130 Steel*, technical reference. Reports normalized fatigue strength ≈ 310 MPa and Q&T fatigue strength ≈ 450 MPa for AISI 4130 chromoly steel.

A Reproducibility

Full code, data, and extended report (21 pages): https://github.com/RudraKhandelwal/ME228

Key files: `data.csv` (NIMS, 437 samples), `Project 2.0.py` (training + CV), `Project 3.0.py` (inverse design), `app.py` (Streamlit GUI), `external_validation.py`, `models/{nc_xgb,c_rf}.pkl`, `models/metadata.json`.

Run order:

```
python "Project 2.0.py"        # train + persist models
python "Project 3.0.py"        # inverse design + Pareto
python external_validation.py  # holdout + matbench + AISI
streamlit run app.py           # interactive GUI
```

B Theoretical Foundations and Bounds (ME228)

Formal theory checks referenced in §8.

B.1 VC Dimension Bookkeeping

$N \geq 10\, d_{VC}$ rule of thumb. Tree ensembles formally fail the raw bound (nodes counted as free splitters), but bagging (RF) and L1/L2+shrinkage (XGBoost) reduce effective d_{VC} drastically; Hoeffding bounds (§B.2) confirm no overfitting.

Table 8: Raw VC-dimension bookkeeping. Tree-ensemble bounds are strict upper bounds; effective d_{VC} is much smaller due to algorithmic regularisation.

Regime	Hypothesis class	N	d_{VC}	N/d_{VC}	Status
NC	Linear regression	389	17	22.9	✓
NC	Ridge ($\alpha = 1$)	389	17	22.9	✓
NC	MLP 1×16	389	289	1.3	low
NC	Decision tree, depth 4	389	272	1.4	low
NC	**XGBoost (chosen):** 311 trees, depth 4	389	4 976	0.08	raw fail*
C	Linear regression	48	8	6.0	low
C	Ridge ($\alpha = 1$)	48	8	6.0	low
C	MLP 1×8	48	73	0.7	raw fail*
C	**RF (chosen):** 238 trees, depth 4	48	3 808	0.01	raw fail*

* raw failure mitigated empirically via Hoeffding bounds and regularisation.

B.2 Hoeffding Generalisation Bound

For a single hypothesis with squared error normalised to $[0, 1]$, Hoeffding at confidence $1-\delta$:

$$P\big(\,|E_{\text{in}} - E_{\text{out}}| > \varepsilon\big) \leq 2\, e^{-2N\varepsilon^2/L^2} \implies \varepsilon = L\sqrt{\frac{\log(2/\delta)}{2N}}$$

With $\delta = 0.05$, $L_{NC} = 681$ MPa, $L_C = 352$ MPa:

Table 9: Hoeffding 95 % generalisation bound vs. measured out-of-fold (OOF) RMSE.

Regime	N	Bound ε (95 %, MPa)	Measured OOF RMSE
NC	389	46.9	20.12 **(43 % of bound)**
C	48	69.0	42.39 **(61 % of bound)**

Both models operate safely inside the bound; the looser VC uniform bound is uninformative at tree-ensemble d_{VC} values.

B.3 Bootstrap Confidence Intervals on RMSE

2 000 percentile-method resamples of OOF residuals per regime:

Table 10: Bootstrap 95 % CI on OOF RMSE. (2 000 resamples)

Regime	Point RMSE (MPa)	Bootstrap Mean	95 % CI
NC	20.12	20.08	[17.14, 23.17]
C	42.39	42.24	[32.51, 51.59]

The C CI width of 19.4 MPa reflects $N = 48$: models between 32.6 and 52.0 MPa cannot be statistically distinguished.